KAIZEN MATHS

# Lesson starter

**Large Data Set Practice**

Graphs & Models

Name

**S1**

Investigate the relationship between rainfall and sunshine hours. State a suitable graph and explain why it is suitable.

**S2**

For rainfall (mm) and sunshine hours, the product moment correlation coefficient is r = -0.78. Interpret this value in context.

**S3**

Use the regression model rainfall = 18.6 + 2.15(wind speed) to predict rainfall (mm) when wind speed (km/h) is 24. Comment on whether the prediction is reliable.

**S4**

Show the distribution of rainfall for all monthly observations. State a suitable graph and explain why it is suitable.



KAIZEN MATHS

Lesson starter

Large Data Set Practice

Graphs & Models

Name

S1

Investigate the relationship between rainfall and sunshine hours. State a suitable graph and explain why it is suitable.

S2

For rainfall (mm) and sunshine hours, the product moment correlation coefficient is r = -0.78. Interpret this value in context.

S3

Use the regression model rainfall = 18.6 + 2.15(wind speed) to predict rainfall (mm) when wind speed (km/h) is 24. Comment on whether the prediction is reliable.

S4

Show the distribution of rainfall for all monthly observations. State a suitable graph and explain why it is suitable.

STA-986D-L3 · S

Show your working

KAIZEN MATHS

# Exit ticket

**Large Data Set Practice**

Graphs & Models

Name

**EA1**

For wind speed (km/h) and rainfall (mm), the product moment correlation coefficient is r = 0.52. Interpret this value in context.

**EA2**

Use the regression model sunshine = 210 - 1.55(rainfall) to predict sunshine hours when rainfall (mm) is 80. Comment on whether the prediction is reliable.



KAIZEN MATHS

Exit ticket

Large Data Set Practice

Graphs & Models

Name

**EB1**

Compare rainfall for coastal and inland towns. State a suitable graph and explain why it is suitable.



KAIZEN MATHS

Exit ticket

Large Data Set Practice

Graphs & Models

Name

EA1

For wind speed (km/h) and rainfall (mm), the product moment correlation coefficient is r = 0.52. Interpret this value in context.

EA2

Use the regression model sunshine = 210 - 1.55(rainfall) to predict sunshine hours when rainfall (mm) is 80. Comment on whether the prediction is reliable.

STA-986D-L3 · EA

Show your working

KAIZEN MATHS

Exit ticket

Large Data Set Practice

Graphs & Models

Name

EB1

Compare rainfall for coastal and inland towns. State a suitable graph and explain why it is suitable.

STA-986D-L3 · EB

Show your working

# Teacher answers & working

**Large Data Set Practice · Graphs & Models**

STA-986D-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

**S1**

Investigate the relationship between rainfall and sunshine hours. State a suitable graph and explain why it is suitable.

**Answer:**

scatter diagram

**Working:**

Identify the variable type and the purpose of the graph.

A suitable graph is: scatter diagram.

Reason: there are two numerical variables and the relationship is the focus.

**S2**

For rainfall (mm) and sunshine hours, the product moment correlation coefficient is r = -0.78. Interpret this value in context.

**Answer:**

fairly strong negative correlation

**Working:**

The sign of r gives the direction. Since r = -0.78, the correlation is negative.

The size of |r| gives strength. |r| = 0.78, so the relationship is fairly strong.

Context: Higher rainfall is associated with fewer sunshine hours.

Remember that correlation alone does not prove causation.

**S3**

Use the regression model rainfall = 18.6 + 2.15(wind speed) to predict rainfall (mm) when wind speed (km/h) is 24. Comment on whether the prediction is reliable.

**Answer:**

Prediction = 70.2

**Working:**

Substitute 24 into the regression equation.

rainfall = 18.6 + 2.15(wind speed).

Prediction = 70.2.

Reliability depends on whether 24 is within the range of observed data and whether the correlation is strong enough for prediction.

# Teacher answers & working

**Large Data Set Practice · Graphs & Models**

STA-986D-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

**S4**

Show the distribution of rainfall for all monthly observations. State a suitable graph and explain why it is suitable.

**Answer:**

histogram

**Working:**

Identify the variable type and the purpose of the graph.

A suitable graph is: histogram.

Reason: rainfall is continuous and grouped intervals show the shape of the distribution.

**EA1**

For wind speed (km/h) and rainfall (mm), the product moment correlation coefficient is r = 0.52. Interpret this value in context.

**Answer:**

moderate positive correlation

**Working:**

The sign of r gives the direction. Since r = 0.52, the correlation is positive.

The size of |r| gives strength. |r| = 0.52, so the relationship is moderate.

Context: Higher wind speed is associated with higher rainfall in this sample.

Remember that correlation alone does not prove causation.

**EA2**

Use the regression model sunshine = 210 - 1.55(rainfall) to predict sunshine hours when rainfall (mm) is 80. Comment on whether the prediction is reliable.

**Answer:**

Prediction = 86

**Working:**

Substitute 80 into the regression equation.

sunshine = 210 - 1.55(rainfall).

Prediction = 86.

Reliability depends on whether 80 is within the range of observed data and whether the correlation is strong enough for prediction.

# Teacher answers & working

**Large Data Set Practice · Graphs & Models**

STA-986D-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

**EB1**

Compare rainfall for coastal and inland towns. State a suitable graph and explain why it is suitable.

**Answer:**

side-by-side box plots

**Working:**

Identify the variable type and the purpose of the graph.

A suitable graph is: side-by-side box plots.

Reason: they show median, spread, skew, and outliers for two groups.